

ABANG OBED

Security Engineer — Security Operations · Offensive Security · Application Security

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PROFESSIONAL SUMMARY

Security engineer with six years of hands-on experience in security operations, vulnerability assessment, application security, detection engineering, and offensive security. My work spans web, API, mobile, network, infrastructure, cloud, and Windows security, including malware analysis and security tooling. I have worked across government and private-sector environments, delivering security assessments, technical reports, and security research.

CORE TECHNICAL SKILLS

Security Operations & Detection: SIEM monitoring, alert triage, threat detection, incident investigation & response, SOC workflows, network & endpoint monitoring, network segmentation, EDR & endpoint hardening

Offensive Security & Assessments: penetration testing, vulnerability assessment, web / API / mobile security assessments, authentication & authorization testing, session management, business logic & injection vulnerabilities, Active Directory exploitation, thick-client enumeration, reverse engineering, malware analysis

Cloud & Infrastructure Security: Microsoft Azure (Entra ID), AWS IAM, network security (MikroTik, Cisco, Huawei NetEngine), Windows internals (Win32 API, Event Tracing for Windows), access control (RBAC, Zero Trust), cloud deployment security

Engineering & Tooling: Python, Node.js, NestJS, Laravel, PostgreSQL, MongoDB, Redis, C#, Swift (iOS), Flutter, React, REST API design, CI/CD pipelines, Terraform (HashiCorp), Git

PROFESSIONAL EXPERIENCE

Lead Security Engineer (Contract)

Kynettic | Remote (Lagos, Nigeria) | July 2026 – Present

- Performed web application, API, and mobile security assessments covering authentication, authorization, session management, business logic, and injection vulnerabilities across client environments.
- Assessed protocol-layer and application security in DeFi systems, investigating swap logic integrity, access control mechanisms, and transaction validation for exploitable flaws.
- Produced structured security assessment reports with severity ratings, clear reproduction steps, and prioritized remediation guidance aligned to the client's risk profile.

Security Operations Specialist (Internship)

National Emergency Management Agency (NEMA-HQ) | Abuja, Nigeria | June 2025 – January 2026

- Monitored and investigated security events in a live government SOC using SIEM and security monitoring workflows, supporting alert triage, threat detection, incident investigation, and response across network and endpoint environments.
- Administered Active Directory and endpoint security infrastructure on HP ProLiant servers using MikroTik and Cisco, enforcing access control policies, endpoint detection and response (EDR), and network segmentation across departments.
- Performed vulnerability assessments of network assets and maintained security posture of Huawei NetEngine CX600 Metro Services Platform; supported emergency communications infrastructure for Geoinformatics (GIS) data-relay coordination and Mission Control Center operations.

Technical Security Writer

ATL Labs (PinkDraconian) | Tesselenderlo, Belgium | June 14 – July 29 2025

- Developed RBAC (Role-Based Access Control) security courseware covering access control architecture, policy design, and enforcement mechanisms for a structured security education programme.
- Conducted research and technical writing on identity management, access control, and security governance, translating concepts into practitioner-focused training material.
- Produced course visualization content for security training aligned to real-world implementation contexts and threat scenarios.

Staff Backend Developer

Phlex IT Events | Remote (London, UK) | April 2024 – July 2025

- Developed REST APIs using NestJS and PostgreSQL for user authentication, OTP verification, session management, and ticket handling at scale.
- Implemented server-side pagination and client caching strategies, improving API responsiveness across datasets exceeding 50,000 attendee records.
- Designed MongoDB ORM architecture and configured CI/CD pipeline for consistent and secure deployment across services.

Sr. Website Developer & System Architect (Contract)

IDE Nigeria (Simdozi LTD, Marblefoods NG, Stonerockers NG, Aoahomes & BigHomes NG) | Abuja, Nigeria | January 2021 – January 2026

- Architected a multi-company operations platform using Node.js, Laravel, Three.js, C#, and Azure with role-based access controls and modular department structure.
- Built mobile-first responsive interfaces using Tailwind CSS, reducing page load times and improving user engagement by 57%.
- Resolved IAM permission misconfigurations and cloud deployment security issues across client environments, improving reliability and security posture.

AWS Cloud Engineer

234coins.net (Freelance, Upwork) | Remote | December 2023

- Optimized AWS IAM service roles to resolve regional permission and upload failures, reducing deployment errors by 50% on the Unity runtime engine.

- Diagnosed cloud CORS misconfigurations and server reliability issues, improving deployment consistency and security.
- Implemented real-time Redis monitoring for token loading and device activity, ensuring accurate gameplay synchronization across regions.

SELECTED RESEARCH & TECHNICAL PROJECTS

GigAfro.com — Freelance Marketplace (Startup, Live Production)

January 2026 | Stack: Flutter (Android), Swift (iOS), React (Web), Apollo GraphQL, WebSockets, Google Places API

- Built and launched a production marketplace connecting freelancers to gigs using real-time geolocation, ranking algorithms, and network-based worker distribution.
- Designed a dispatch system converting user intent into real-world gig assignments across Android, iOS, and web clients in a live production environment.

Conversational AI for Dementia Care

NCAIR Hackathon Winner, \$1,000 — September 2025 | Stack: Swift (iOS), React, Wandb, Emotion Recognition ML (pet-emotion-k8t5, empathy-dementia), Terraform & HashiCorp

- Designed backend services for conversation state management, intent handling, and context-aware LLM-powered dialogue; integrated emotion recognition to adapt responses to inferred cognitive state.
- Built a full-stack dialogue system for real-time dementia support across iOS and web clients using Swift, React, and Node.js with Terraform infrastructure.

LiDAR Spatial Assistant for Blind Users

Gemini Hackathon Special Recognition — February 2026 | Stack: React Native (ARKit, LiDAR), YOLOv9, ARCore, real-time 3D mesh, TTS audio feedback

- Developed an iOS application using device LiDAR for real-time 3D room scanning and object detection; integrated YOLOv9 and ARCore for spatial awareness and audio navigation cues for blind users.

Autobot OS — Custom Python Operating System Runtime

September 2023 | Stack: Python, Win32 API, Windows Internals, Event Tracing for Windows, Capstone, Keystone

- Built tooling to observe execution paths and collect endpoint telemetry, supporting research into how code-level changes affect Windows security visibility and EDR resilience.

AZURE-STACK-USSD

October 2023 | Stack: Azure Stack Hub, Africa's Talking API, USSD, Entra

- Built a scalable USSD service on Azure Stack Hub, integrating Africa's Talking API to provide enterprise mobile access to USSD services.

PUBLICATIONS

[1] Abang, O. (2026). Design and Implementation of a Secure Telemetry-Driven Code Mutation Framework for Controlled Endpoint Detection and Response Resilience Testing in Windows Environments. [B.Tech Thesis, Federal University of Technology, Minna]

[2] Abang, O. & Seun Ajayi. (2025). Conversational Artificial Intelligence for Dementia Care: Emotion-Aware Dialogue Design and System Architecture. [Technical Report, National Centre for Artificial Intelligence and Robotics (NCAIR), Abuja]

EDUCATION & CERTIFICATIONS

B.Tech in Cyber Security Science

Federal University of Technology, Minna | Minna-Niger, Nigeria | 2021 – 2026

Focused on operating systems, Windows internals, endpoint security, and telemetry research; developed custom Windows runtime and security tooling using Python and the Win32 API to study process execution, code mutation, and EDR telemetry resilience.

Certified Penetration Testing Specialist (CPTS)

HackTheBox EU — Credential ID: HTBCERT-2508B8ABE8 | February 2024

Active Directory exploitation, penetration testing, vulnerability assessment, thick-client enumeration, reverse engineering, security reporting and documentation.

HONORS & AWARDS

- National Center for Artificial Intelligence & Robotics (NCAIR) Grant Recipient (2025) — Awarded \$1k grant to develop the Dementia ChatBot.
- CodeSandbox Security Vulnerability (2024) — Officially credited for reporting a high-severity security issue on CodeSandbox's GitHub Security Advisory program (GHSA-5jw5-g7mr-h26r).
- X (Twitter) Security Vulnerability (2023) — Improper access control vulnerability, classified as critical severity on HackerOne with a \$2K bounty; CyberSafe HQ HackTheBox 1st Place Winner.

TALKS & COMMUNITY IMPACT

- CyberNexus Starter, Federal University of Technology, Minna (2026) — Threat detection approaches for iOS application architecture, malware analysis, reverse engineering, and antimalware techniques.
- NaCss Cybersecurity Awareness Month, Al-Amin Secondary School (2023) — Wireless phishing attacks and awareness using Airgeddon.
- Campus Lead, CSEAN (Cyber Security Experts Association of Nigeria) Student Chapter (2024 – 2025) — Led campus cybersecurity initiatives, student engagement, awareness, and technical learning activities.
- Graphic Designer & Video Editor, NaCss Cybersecurity Awareness Month (2022) — Produced awareness content on Zero Trust.